Assignment 6

Total = 10 Marks

Multiple Choice Questions

Question 1: In a Simple Linear Regression model, what assumption is made regarding the residuals (ϵ_i)?

(a) They follow a uniform distribution with a mean of one.
(b) They are dependent on the variable X and skewed.
(c) They are independent and normally distributed with a mean of zero and a variance of σ^2.
(d) They continuously increase as the value of X increases.

[1 *Mark*]

Correct answer: (c) They are independent and normally distributed with a mean of zero and a variance of σ^2.

Detailed Solution:

For fitting a linear regression, following assumptions are made:

i. The relationship between the X and mean of Y is a straight line. Hence, for a specific value of X and Y (represented by x_i and y_i, respectively), the regression model is given by:

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

the random variables y_i are independently normally distributed with a mean of $\alpha + \beta x_i$ and variance equal to standard error (represented by σ^2), for $i = \{1, 2, \dots, n\}$.

ii. The residuals (ε_i) are independent and normally distributed with a mean of zero and the variance of σ^2.

Question 2: What is the theoretical range of the coefficient of determination (R^2)?

(a) -1 to 1.
(b) $-\infty$ to 1.
(c) 0 to 1.
(d) 0 to ∞.

[1 *Mark*]

Correct answer: (c) 0 to 1.

Detailed Solution:

The adequacy of regression model can be expressed as ratio of variability explained by the regression model $\sum(\hat{y}_i - \bar{Y})^2$ and total variability in observed dependent variable $\sum(y_i - \bar{Y})^2$. The ratio is called coefficient of determination and represented as r^2 or R^2.

The ratio of variability explained by the regression model can never be greater than total variability of the dependent variable. Hence, the coefficient of determination ranges between 0 and 1. Closer the R^2 to 1, the better the model is.

Question 3: When analyzing data where the dependent variable is modeled by $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_p x^p + \epsilon$, how can the coefficients be solved using least squares?

(a) By utilizing logarithmic transformations to force the equation into a simple linear state.
(b) By considering different powers of the independent variable as separate independent variables and applying the concept of multiple linear regression.
(c) By applying reciprocal transformations to the dependent variable.
(d) None of these

[1 *Mark*]

Correct answer: (b) By considering different powers of the independent variable as separate independent variables and applying the concept of multiple linear regression.

Detailed Solution:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_p x^p + \varepsilon$$

The given equation is an example of polynomial fit between Y and X. It represents a case where the functional form of the regression Y on X is not transformable to linear regression. The corresponding coefficients can also be calculated by the method of least squares as discussed previously. The coefficients of fit (β_i) can be obtained by considering different powers of independent variable as separate independent variable and using the concept of multiple linear regression.

Question 4: During an infiltration experiment, the initial soil moisture content (X, in %) and the corresponding infiltration capacity (Y, in cm/hr) are measured at different locations in an agricultural field. It is assumed that the relationship between the initial soil moisture content and infiltration capacity follows the reciprocal model: $y = \frac{1}{\alpha + \beta x}$. Determine the values of the coefficients α and β using Method of Least Squares.

X	13.50	17.30	30.00	14.20	15.40	15.37
Y	9.51	12.07	15.84	12.65	8.73	14.34

(a) $\alpha = 0.01,\ \beta = 0.02$
(b) $\alpha = 0.09,\ \beta = 0.05$
(c) $\alpha = 0.12,\ \beta = 0.002$
(d) $\alpha = 0.19,\ \beta = 0.005$

[4 *Mark*]

Correct answer: (c) $\boldsymbol{\alpha = 0.12,\ \beta = 0.002}$

Detailed Solution:

The general form of a simple reciprocal function is given by:

$$y = \frac{1}{\alpha + \beta x}$$

Because this equation is non-linear with respect to its parameters, standard linear regression cannot be applied directly. It must be transformed into a linear format.

By taking the reciprocal of both sides,

$$\frac{1}{y} = \alpha + \beta x$$

Let $Y' = \frac{1}{y}$. The equation now takes the standard linear form:

$$Y' = \alpha + \beta x$$

where, Y' is the transformed dependent variable, X remains the independent variable, α is the intercept of the transformed linear model and β is the slope of the transformed linear model.

To use the Method of Least Squares, the summations of X, Y', X^2, and XY' need to be calculated. The table outlines the necessary calculations.

Sl. No.	y'_i	x_i	x_i^2	$x_i y'_i$
1	0.105	13.500	182.250	1.420
2	0.083	17.300	299.290	1.433
3	0.063	30.000	900.000	1.894
4	0.079	14.200	201.640	1.123
5	0.115	15.400	237.160	1.764
6	0.070	15.370	236.237	1.072
Sum	0.514	105.770	2056.577	8.705

The parameter β can be calculated as: $\beta = \frac{S_{xy}}{S_{xx}}$

$S_{xy} = \sum_{i=1}^{n} x_i y_i - \frac{\sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n} = (8.705) - \big((105.770) \times 0.514\big)/6 = -0.356$

$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} = 2056.577 - (105.770)^2/6 = 192.028$

Thus, $\beta = \frac{S_{xy}}{S_{xx}} = \frac{-0.356}{192.028} = -0.0018 \approx 0.02$

The parameter α is calculated as:

$\alpha = \frac{1}{n}(\sum_{i=1}^{n} y_i - b \sum_{i=1}^{n} x_i) = (0.514 - (-0.002) \times (105.77))/6 = 0.121 \approx 0.12$

The final curvilinear regression model would be written as:

$$\frac{1}{y} = 0.12 - 0.002x \text{ or } y = \frac{1}{0.12 - 0.002x}$$

Question 5: A multiple linear regression model is written as $E(Y|X_1,\ X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.

For a dataset, the following matrices have been obtained:

$$(X^T X)^{-1} = \begin{bmatrix} 0.90 & -0.20 & -0.03 \\ -0.20 & 0.06 & 0 \\ -0.03 & 0 & 0.02 \end{bmatrix}$$

$$(X^T Y) = \begin{bmatrix} 200 \\ 500 \\ 900 \end{bmatrix}$$

Determine the value of β_0, β_1, β_2 and $E(Y|13,\ 26)$.

(a) $\beta_0 = 53, \beta_1 = -10, \beta_2 = 12, E(Y|13, 26) = 235$
(b) $\beta_0 = -10, \beta_1 = 53, \beta_2 = 12, E(Y|13,\ 26) = 991$
(c) $\beta_0 = 53, \beta_1 = 12, \beta_2 = -10, E(Y|13,\ 26) = -51$
(d) $\beta_0 = -53, \beta_1 = -10, \beta_2 = 12, E(Y|13,\ 26) = 129$

[3 *Mark*]

Correct Answer: (a) $\boldsymbol{\beta_0 = 53, \beta_1 = -10, \beta_2 = 12, E(Y|13,\ 26) = 235}$

Detailed Solution:

Given:

$$(X^T X)^{-1} = \begin{bmatrix} 0.90 & -0.20 & -0.03 \\ -0.20 & 0.06 & 0 \\ -0.03 & 0 & 0.02 \end{bmatrix}$$

$$(X^T Y) = \begin{bmatrix} 200 \\ 500 \\ 900 \end{bmatrix}$$

Therefore, $\hat{\beta} = (X^T X)^{-1} X^T Y = \begin{bmatrix} 53 \\ -10 \\ 12 \end{bmatrix}$

Hence, the relationship is given by :

$$E(Y|X_1,\ X_2) = 53 - 10X_1 + 12X_2$$

Also, $E(Y|13,\ 26) = 53 - 10 \times 13 + 12 \times 26 = 53 - 130 + 312 = 235$